



OPEN A hybrid feature selection framework for dimensionality-optimized customer churn prediction

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Customer churn is a major problem in the fiercely competitive telecom industry. It directly influences the profitability of a company and its future in the industry. There have already been successful applications of machine learning and deep learning (ML/DL) techniques to detect client churn. However, in certain situations, ML/DL-based algorithms fall short of providing encouraging outcomes when it comes to classifying customer churn. The use of machine learning classifiers and conventional feature encoding techniques in earlier studies on customer turnover estimation produced promising predictions. But the role of the optimized feature set remains important for improving efficiency and reducing feature dimension. This research proposes a modular churn prediction framework that integrates data preparation, feature optimization through hybrid feature selection, class balancing, and predictive modeling into a unified pipeline. This is done through a Multi-stage Hybrid Feature Selection (MHFS) technique that combines Mutual Information (MI) for preliminary relevance filtering, Recursive Feature Elimination (RFE) for iterative refinement, Boruta for rigorous variable validation, and Parallel Simulated Annealing (SA) for global optimization. Furthermore, a 5 × 3 nested stratified cross-validation is used to guarantee objective model evaluation where feature selection and SMOTE are performed exclusively within inner training folds, while outer folds remain completely unseen for performance assessment. Multiple ML algorithms are used for training, including Random Forest, XGBoost, LightGBM, SVM, Extra Trees, and Logistic Regression, along with Tabular Neural Network. The framework is evaluated using two comprehensive telecommunications datasets, Cell2Cell and IBM Telco. It will ensure the efficacy of the proposed framework across different churn settings datasets, which is then combined with nested cross-validation on the training set. The experimental results show that the proposed hybrid framework achieves competitive predictive performance while substantially reducing feature dimensionality (from 28 to 16 features in IBM Telco and from 57 to 25 features in Cell2Cell). Using XGBoost, the framework achieves an accuracy of 81.60% on IBM telco and 77.52% on the cell2cell dataset. These results show that the proposed framework provides competitive performance while significantly reducing feature dimensions.

Keywords Customer churn, Feature selection, Simulated annealing, Recursive feature elimination, Mutual information

Customer churn, also known as customer attrition, is the process of consumers discontinuing a subscription to a company's product or service. A customer churn rate may be used to figure out this important business number that shows the loyalty and happiness of the customer. As markets mature, competition becomes fiercer, and the cost of getting new customers is far higher than the cost of keeping old ones, so customer attrition has become more important than ever. In this situation, reducing turnover has become an important part of business strategy. It seeks to determine when, how, and why clients cease using a service¹.

This is especially challenging in the telecommunications industry as the service providers are facing more competition, lower profit margins, and higher customer expectations for both price and quality. Many companies

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have moved from getting new customers to keeping their current ones, realizing that churn prediction models are necessary to stay ahead of the competition. Client attrition is still a big problem, even though client relationship methods have improved over the years. Predictive analysis alone cannot fully explain the psychological and behavioral factors that affect client choices².

Previous studies have employed statistical and probabilistic methods to identify clients with the highest probability of attrition³. But these studies often ignored the essential behavioral, economic, and service-related elements that contribute to attrition, which led to methods that lacked interpretability and contextual understanding⁴. Recent studies have tried to rectify this issue by employing comprehensive churn prediction models that integrate both predictive precision and transparency^{4,5}. The growth of bundled telecommunications services, which include mobile, internet, and television, has made churn dynamics more complicated. Bundling techniques aim to boost customer loyalty by making it hard and expensive to switch⁶. Even though these techniques are common, there isn't much research on how people who use bundled services behave when they leave, especially in developing nations. Fig. 1 presents different approaches for customer churn prediction.

Machine learning has been utilized in the past for customer churn prediction⁷. Recent studies show the effectiveness of ML in various areas, such as autism and diabetes detection^{8,9}, fake news identification¹⁰, and sentiment analysis^{11,12}. This research is motivated by these studies to analyze the application of advanced Machine Learning Algorithms (MLAs) to enhance churn prediction. Even though there have been a lot of advances in M-based churn prediction, they still struggle with class imbalance and redundant features in customer data. It creates model confusion and reduces the accuracy of the forecasts. The telecom sector is going through a big transformation driven by 5G deployment and IoT integration, resulting in high-dimensional and heterogeneous customer data. Systematic feature optimization is important because redundant and noisy attributes can reduce generalization performance and increase operational cost in the long term. The proposed hybrid multi-stage feature selection framework addresses these challenges by integrating filter, wrapper, embedded, and stochastic optimization strategies within a statistically validated nested cross-validation design, thereby emphasizing dimensional efficiency with competitive predictive performance. This research is driven by the need to develop a low-dimensional hybrid feature selection framework that helps competitive churn forecasting while minimizing feature dimensions and enhancing generalization across datasets. This strategy would give telecom companies useful information that would help them lower churn and keep long-term relationships with clients^{5,13}.

The main study contributions are:

1. Proposing a modular churn prediction framework that integrates data preprocessing, feature selection, class balancing, and advanced modeling in a unified pipeline.
2. Introduction of a hybrid multi-stage feature selection framework to derive an optimal subset of predictive features by integrating Mutual Information (MI) for initial relevance filtering, Recursive Feature Elimination (RFE) for iterative refinement, Boruta for robust variable validation, and Parallel Simulated Annealing (SA) for global optimization of the selected feature subset.
3. The integration of multiple ML and DL algorithms within a unified framework of hybrid feature selection provides competitive performance while reducing dimensionality for churn prediction.

The remainder of the study is structured as follows: the following Section "Methodology" discusses the literature review. Section 3 explains the technique used, while Section "Results and discussion" describes the experiments, findings, and assessment methods. The final section concludes the study.

Literature review

The telecom industry has done a lot of research on customer churn because it is an important factor that affects how much money a business makes and how long its relationships with customers last. The initial research mostly focused on making predictive models that use probabilistic and decision-tree-based classifiers to find clients who are likely to cancel their subscriptions^{3,4}. These models focused on predicting performance by using metrics like accuracy, precision, and recall. This showed how useful data mining and machine learning approaches might be for churn monitoring.

New research in ML has made it easier to predict how many customers will leave the telecoms industry. The addition of ML in the areas of medical NLP^[14], student performance analysis^[15], and spambot detection

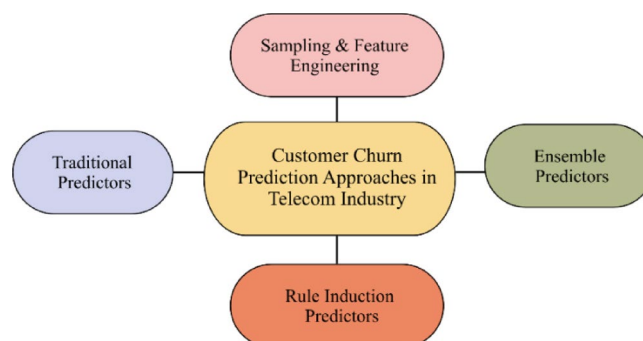


Fig. 1. Customer churn prediction approaches in the telecom industry.

[16] allows the researchers to modify these techniques for churn prediction. Researchers have looked into a number of ML and DL algorithms to see how well they can model customer retention and attrition. These techniques look for both linear and non-linear connections in customer data. They fix problems like class imbalance, noise, and feature correlation while also making predictions more accurate and easier to understand.

Logistic Regression is commonly used for classifying churn since it is easy to use and understand. Research [17–19] integrated Logistic Regression with Decision Tree models to categorize clients and subsequently forecast turnover. These models had accuracies between 74 and 85%, which means they worked well to tell the difference between people who were going to leave and those who weren’t. Logistic regression was very useful for figuring out how things like how long a consumer has been with a service, what kind of service they subscribe to, and how they pay affect churn. But because it was linear, the model had trouble capturing complicated, non-linear relationships in big datasets [20,21].

Reference [7] provides a comprehensive analysis of customer attrition prediction in the telecom sector. The study shows numerous attributes employed by a substantial number of article reviewers to develop a customer churn prediction model. Reference [22] conducted a comparative analysis of customer churn prediction in the telecommunications sector utilizing established ML methodologies. The experimental results show that the ensemble learning algorithms like Adaboost and XGBoost classifiers are the most accurate compared to other methods. They got an AUC value of 84 percent for the customer attrition problem, which is better than other algorithms. An enhanced customer attrition model for credit card systems utilizing supervised learning and rough clustering techniques is provided in reference [23]. Authors of reference [24] employed Naïve Bayes for churn prediction among telecommunication consumers, emphasizing the difficulties of acquiring new clients rather than managing existing ones. A model for analyzing client behavior was established, accompanied by enhanced predictive accuracy. In reference [25], the authors emphasized churn prediction in the telecommunications sector with a Just-in-Time technique. It addressed several issues with customers’ intentions to switch from one service provider to the next. In this scenario, many churn prediction methodologies are employed by practitioners to address challenges associated with client retention.

Deep learning has transformed churn prediction research by facilitating automatic feature extraction from the data. Convolutional Neural Networks (CNN) have shown exceptional efficacy in acquiring non-linear and hierarchical representations of data [26–28]. Researchers have compared CNN with classic ML methods and have shown that deep learning models do better than most ML algorithms in terms of performance. CNN-based churn prediction frameworks attained accuracy rates beyond 95%, with F-scores of 92% and AUC values nearing 0.99. Additionally, resampling approaches were used to fix data imbalance, which made the model more generalizable. Previous studies have also focused on optimizing CNN topologies to improve computational efficiency and to maintain the reliability of prediction [29].

Recent studies have shown that there is a need to forecast customer turnover so that the profitability and competitiveness can be improved in the telecom sector. Reference [30] used Fisher discriminant equations and logistic regression to create a churn prediction model utilizing data from three large Chinese telecom firms. The model was able to predict churn with a high accuracy of 93.94%. Another study investigated the application of advanced machine learning methodologies and optimization algorithms, combined with feature selection and undersampling techniques, to improve the performance of churn prediction [31]. The Random Forest model improved by Grid Search (RF-GS-LR) worked better than any other model, with accuracy rates of up to 99%. It shows that combining statistical and machine learning methods can greatly help telecom businesses find consumers who are at risk of leaving and come up with good ways to keep them. Alternative methodologies, like rough set theory, have been employed for rule-based churn classification. Reference [32] used different rule generation methodologies like genetic, exhaustive, and LEM2 algorithms, to create choice rules. When combined with genetic algorithms (GA) for rule optimization, these methods exhibited a significant improvement in performance, as measured by F-measure, accuracy, precision, and recall. Table 1 provides the latest literature analysis of Churn prediction.

Methodology

This section goes into depth on the proposed churn prediction framework, which includes a discussion on all components and steps for preparing and modeling the data. To make churn prediction more accurate and reliable, the framework has a number of interconnected parts, each of which serves a different analytical purpose in the predictive pipeline.

Ref	Year	Objective/focus	Methodology/algorithms	Key findings
33	2024	To predict telecom customer churn and identify key churn factors	Used classification and clustering with XGBoost, Random Forest, KNN, SVM, and Decision Trees	XGBoost and Random Forest achieved the highest accuracy, precision, F1-score, and recall
34	2024	To design a profit-oriented and interpretable churn prediction framework that maximizes business value through ensemble learning	Developed a Profit-Driven Weighted Classifier integrating multiple ensemble members using Artificial Hummingbird Optimization (AHO). Interpretability enhanced through SHAP	The proposed classifier improved profits and interpretability compared to traditional models. Demonstrated effectiveness in balancing predictive power with explainability
35	2024	To enhance churn prediction by integrating temporal, spatial, and contextual feature extraction using hybrid neural networks	Used ADASYN sampling for balance; hybrid architecture combining CNN + BiLSTM + Multi-Head Self-Attention	Achieved > 92% precision across datasets; CCP-Net improved performance by 1–3% over other hybrid models
36	2025	To compare ensemble and classical ML models for churn prediction	Compared Ensemble Models vs. classical classifiers using confusion matrices and ROC curves	Ensemble models outperformed classical classifiers on all performance metrics

Table 1. Latest literature analysis of churn prediction.

Proposed framework

In this research, a complete framework is introduced for predicting telecom customer churn, structured as a modular framework that guarantees excellent predictive accuracy and scalability. This framework brings together a number of activities that are needed for effective churn classification, including data partitioning, preprocessing data, choosing features, and testing models. We use two publicly available datasets in our system, including Cell2Cell and IBM telecom churn, as explained in Sect. 3.2. Many datasets include uneven distributions and irregularities that require strict preprocessing methods to make the data better and more representative.

To ensure unbiased model evaluation and prevent data leakage, a 5×3 nested stratified cross-validation framework was adopted. The dataset was first partitioned into five outer stratified folds, where each outer test fold remained completely unseen during feature optimization, preprocessing, and model training. Initial data preparation included data cleaning, categorical encoding, and missing value imputation. Within each outer training partition, hybrid feature selection was performed to identify the most informative predictors of customer churn while reducing model complexity, as detailed in Sect. 3.4. The feature selection stages including Mutual Information (MI), Recursive Feature Elimination (RFE), Boruta, and Simulated Annealing were executed exclusively using training data within the inner cross-validation procedure to ensure strict isolation of the outer test fold and eliminate information leakage. The optimal feature subset was selected based on the mean inner-fold F1-score and subsequently applied unchanged to the corresponding outer test fold. After feature selection, feature standardization parameters were estimated using only the outer training data and applied to both the training and test partitions. Class imbalance was then addressed using SMOTE-based resampling applied exclusively to the scaled outer training data, ensuring balanced model learning while maintaining complete separation from unseen test samples.

The churn classification component, which is explained in Sect. 3.6, were trained using the balanced outer training data and evaluated on the untouched outer test folds. Performance metrics obtained across all outer folds were aggregated to provide statistically reliable estimates of model generalization performance. Fig. 2 shows how the proposed framework works as it turns raw telecom data into actionable churn insights from start to finish.

Dataset

The data used in the current study comprised two comprehensive online available repository datasets, namely the International Business Machines (IBM) Telco and Cell2Cell. Both of these datasets are unbalanced, as shown in Fig. 3. To deal with this issue, we utilize resampling as discussed in Sect. 3.6.

IBM Telco: The IBM Corporation provided this dataset, which was gained from the Kaggle Data Platform. The data contained 7,043 samples with 21 attributes that depicted various types of customers, service attributes, and account information. The dataset was not balanced, with 1,869 customers who had left and 5,174 customers who remained. All customers were registered in California, USA.

Cell2Cell: The dataset was from the Teradata Center for CRM at Duke University in the USA. A total of 71,047 samples with 58 attributes were involved that described customer demographics, value-added services, payment methods, and patterns of use. Customers who have left were 14,257, while 35,519 customers remain.

We consider a binary classification problem for customer churn. Let the original dataset be as shown in Eq. 1:

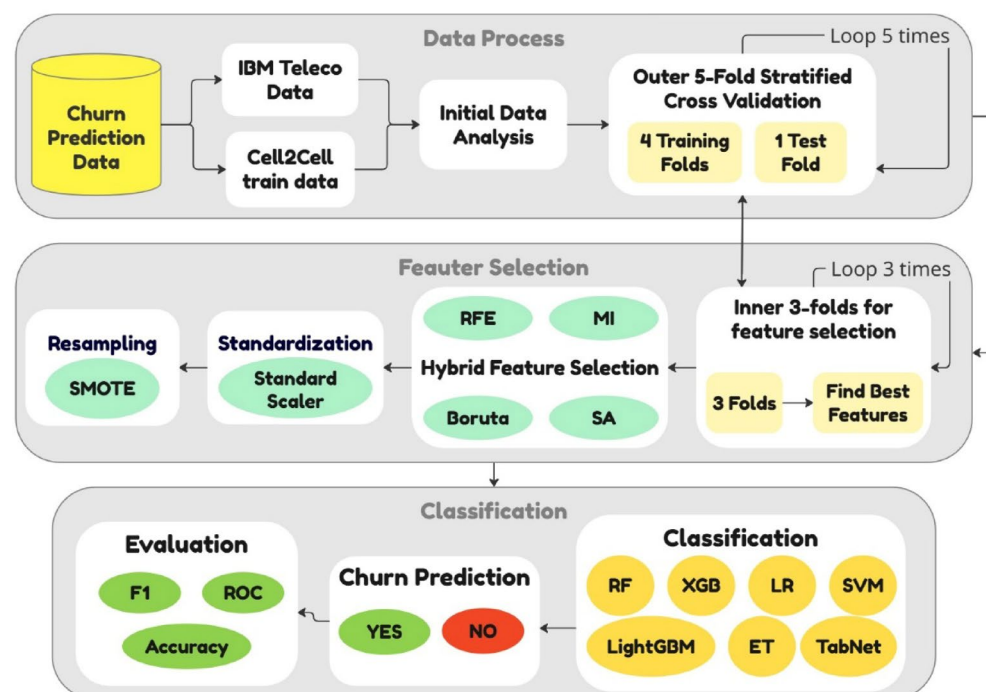


Fig. 2. Hybrid feature selection-based churn prediction framework.

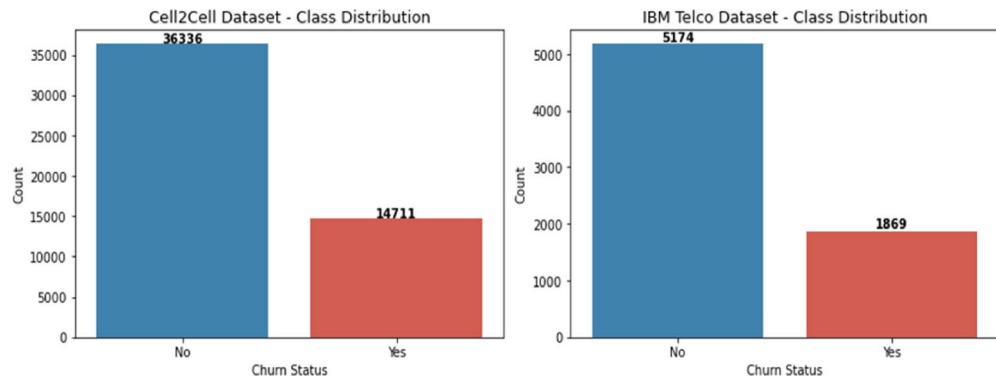


Fig. 3. Class distribution for both datasets.

$$D = \{(x_i, y_i)\}_{i=1}^N, \quad (1)$$

where $x_i \in X$ is the feature vector for the customer i and $y_i \in \{0, 1\}$ is the churn label (1 = churn, 0 = retained). Two datasets are used independently and for combined experiments: IBM Telco Customer Churn and Cell2Cell churn. Each x_i may include numeric features (e.g., tenure, monthly charges) and categorical features (e.g., contract type, payment method). The dataset often exhibits class imbalance. The objective is to learn a classifier $f: X \rightarrow [0, 1]$ such that the predicted probability $p_i = f(x_i)$ produces accurate binary decisions $y_i = \prod (y_i \geq \tau)$ for a threshold τ (commonly 0.5), while addressing class imbalance and high-dimensionality via resampling and hybrid feature selection.

Data analysis

Pre-processing was carried out to ensure consistency in the data sets and the model training before the data was fed to the classification module. The Cell2Cell and IBM Telco Customer Churn data sets were pre-processed in a similar manner to ensure consistency. The initial step was to identify and eliminate duplicate values in the data sets, as they could influence the model training. The framework is more reliable if duplicates are removed to ensure that each data piece is associated with a unique customer. The churn variable was converted to binary form to ensure consistency in the data sets. The “Churn” column in the Cell2Cell and IBM Telco data sets was updated to “1” and “0” for customers who churn and those who have not churned, respectively. The missing values were handled with care to ensure that vital information was not lost. For this, the columns containing excessive missing values were removed if they were greater than 30% of the total entries to minimize the “noise” factor during the training process. Some of the features were categorical, which needed to be converted to numerical values to fit the framework. For this, the Label Encoder was used to transform the categorical attributes to corresponding integer values. One-Hot Encoding was used for features with multiple categories to avoid ordinal bias. This was done to ensure that the ML algorithms precisely understand the interrelations between the features without getting the categories wrong. Z-Score normalization was used to ensure uniformity among the continuous numerical values to avoid bias towards the attributes with larger scales. In this, all the characteristics were standardized to a mean of zero and a standard deviation of one. This is necessary for algorithms like SVM and Neural Networks, which are sensitive to the scales.

Hybrid feature selection

In this research, the Hybrid Multi-stage Feature Selection approach is proposed for the selection of an ideal set of features that can enhance the performance of the churn prediction model while reducing false positives. Feature selection is considered an important issue while dealing with high-dimensional data in the context of telecom churn prediction. It can make the model more generalizable and easier to compute. The proposed approach is based on the hybridization of the best characteristics of the filter, wrapper, and metaheuristic approaches.

In the first phase, Mutual Information (MI) is used for filter-based selection, which examines the relationship between every input attribute and the target class and ranks them according to their effectiveness in reducing uncertainty around churn prediction. We eliminate non-essential attributes early by setting the limit to top 40 features. It decreases the dimensionality of the dataset and ensures that the features exhibiting a robust non-linear correlation with the churn outcome are kept for further testing. The MI score for each feature f_j with respect to target ‘Y’ is expressed as shown in Eq. 2:

$$MI(f_j, Y) = \sum_{f_j} \cdot \sum_Y p(f_j, Y) \log \frac{p(f_j, Y)}{p(f_j)p(Y)} \quad (2)$$

In the second step, Recursive Feature Elimination (RFE) is employed. This is a wrapper-based technique. RFE works by repeatedly training the model, which is normally a Random Forest, while repeatedly removing the least important features according to the coefficient weights or the feature importance. This process is continued until the desired number of features is found by cross-validation. This allows us to view the relationship between

multiple attributes, which cannot always be done by the univariate filters. This is an improvement over the subset generated in the previous MI. The parameters were set as $n_estimators=100$ and $random\ state=42$. The selection criterion minimizes classification error 'E' over subsets 'S', where $M(S)$ denotes the model trained on subset 'S' as shown in Eq. 3.

$$S_{RFE} = \arg \min_{S \subseteq F} E(M(S)) \quad (3)$$

In the third step, Boruta is used as a strong validation tool to make sure that the qualities kept from earlier stages are still useful. Boruta is a comprehensive feature selection method that aims to identify even subtle yet persistent indicators of churn. It works by comparing the importance of real characteristics to that of randomly chosen "shadow features." Features that consistently exceed these shadow variables throughout several iterations are considered meaningful, but those that do not are discarded. This makes sure that the resultant subset of features is strong and dependable. The base estimator was selected as Random Forest, max depth as 6 and random state as 42. A feature f_j is retained if:

$$I f_j > \max(I_{shadow}) + \epsilon \quad (4)$$

where $I f_j$ is the mean importance of the real feature, I_{shadow} is the maximum shadow importance, and ϵ is a small tolerance constant.

In the final step, Parallel Simulated Annealing (SA) is applied as a metaheuristic search method for optimization purposes. In the previous steps, a statistical and model-based approach has been applied, but in this step, a random global search is applied to find the optimal feature set for the final prediction goal. The algorithm begins with a set of characteristics, and then elements are added or removed to explore new possibilities in the neighborhood to find the optimal set of characteristics. By applying a random search, it is possible to surpass local minima and find the optimal set of characteristics on a global scale. By applying the simulated annealing method in parallel on multiple random seeds, convergence can be accelerated. The purpose of this step is to find a set of characteristics to improve the overall efficacy of classification, combining performance indicators such as accuracy, F1-score, and AUC. Equation 5 shows the acceptance probability 'P' of transitioning from one feature subset S_i to another S_j , where $\Delta E = E(S_j) - E(S_i)$ represents the change in evaluation metric (e.g., error or loss), and 'T' denotes the temperature parameter controlling exploration. The Random Forest classifier employs $class_weight = \text{"balanced_subsample"}$. This modification ensures that feature subset evaluation is not biased toward the majority class.

$$P = \exp\left(-\frac{\Delta E}{T}\right) \quad (5)$$

The main benefit of the hybrid design is that it combines several methods: Filter techniques make things go more smoothly, wrapper methods improve the model, and metaheuristic optimization makes sure that the whole feature space is explored. This multi-stage strategy helps avoid overfitting, improves generalization across datasets, and gives a consistent set of features that are reliable even when class distributions or sampling methods change. The method works well to find the most important factors that lead to churn, such as how long a client has been with the company, what sort of contract they have, how often they use the service, and how they engage with customer care. These traits make classification accurate and give telecom companies useful data about how customers act, which helps them to produce data-driven plans to keep customers. The final optimal feature subset is derived as shown in Eq. 6:

$$F = SA(Boruta(RFE(MI(F)))) \quad (6)$$

In Algorithm 1, the complete dataset is denoted as $D = \{(X, y)\}$, where $X \in \mathbb{R}^{n \times d}$ represents the feature matrix with n samples and d features, and $y \in \{0, 1\}^n$ denotes the corresponding binary churn labels. The nested cross-validation framework consists of $K = 5$ outer folds and $L = 3$ inner folds. For each outer iteration $k \in \{1, \dots, K\}$, the dataset is partitioned into training and testing subsets, denoted as $D_{train}^{(k)}$ and $D_{test}^{(k)}$, respectively. Within the outer training partition, feature selection configurations are represented by $C \in \{C_0, C_1, C_2, C_3, C_4\}$, where $C_0 = \text{All features}$, $C_1 = \text{Top-40 features via Mutual Information}$, $C_2 = MI \cup RFE$, $C_3 = MI \cup RFE \cup Boruta$, $C_4 = \text{Apply Simulated Annealing on } C_3$. The optimal configuration selected within outer fold k is denoted by C_k^* , and the resulting feature subset is represented by $S_k \subseteq \{1, \dots, d\}$. During the Simulated Annealing stage, feature subsets are encoded as a binary mask $m \in \{0, 1\}^d$, where $m_j = 1$ indicates that feature j is selected. The objective function $f(m)$ corresponds to the mean inner-fold F1-score obtained using the Random Forest classifier on the selected feature subset. Across outer folds, performance metrics are aggregated using the mean μ and standard deviation σ . For statistical comparison, McNemar's test utilizes the discordant pair counts b and c , representing instances misclassified differently by two competing models.

Input: Dataset $D = \{(X, y)\}$,

classifiers {RF, XGB, SVM, ET, LightGBM, LR, TabNet}

Outer folds $K = 5$, Inner folds $L = 3$

- i. Preprocess Data
Encode categorical variables, impute missing values, and convert the target to binary.
- ii. Outer Stratified Cross-Validation
For each outer fold $k = 1, \dots, 5$:

Split: $D \rightarrow D_{\text{train}}^{(k)}, D_{\text{test}}^{(k)}$

- iii. Inner Cross-Validation (MI $\rightarrow$ RFE $\rightarrow$ Boruta)

Within $D_{\text{train}}^{(k)}$:

For each configuration $C \in \{C_0, C_1, C_2, C_3, C_4\}$:

Perform feature selection strictly on inner training folds:

$$S_{k,C} = FS_C(X_{\text{train-inner}}^{(k)}, y_{\text{train-inner}}^{(k)})$$

Evaluate each configuration using inner CV mean F1-score:

$$\bar{F}1_C = \frac{1}{L} \sum_{l=1}^L F1_C^{(l)}$$

Select optimal configuration: $C_k^* = \arg \max_C \bar{F}1_C$

Final feature subset for outer fold: $S_k = S_{k,C_k^*}$

MI: Top 40 features in descending order

RFE: $n_{\text{estimators}} = 100$, random state = 42

Boruta: Base estimator = RF, max depth = 6, random_state = 42

- iv. Simulated Annealing (Executed within Inner Loop Only)

Initialize binary mask: $m \in \{0,1\}^d$

Objective function (validated via inner CV):

$$f(m) = \text{Mean F1}_{\text{inner-CV}}(\text{RF}(X_m), y)$$

Where RFbalanced = RF(class_weight = "balanced_subsample")

Acceptance criterion: m' accepted if $f(m') > f(m)$

Or $\exp\left(\frac{f(m') - f(m)}{T}\right)$

Temperature update: $T \leftarrow \alpha T$

- v. Standardization $X' = \frac{X - \mu_{\text{train}}}{\sigma_{\text{train}}}$

Scaling parameters computed on the outer training set only and used to transform testing set.

- vi. Class Balancing (Outer Training Only)

Apply SMOTE only on selected features of the outer training set:

$$(X_{\text{bal}}^{(k)}, y_{\text{bal}}^{(k)}) = \text{SMOTE}(X_{\text{train}}^{(k)}[S_k])$$

- vii. Model Training & Testing (Outer Fold)

For each model $M \in \{\text{RF}, \text{XGB}, \text{SVM}, \text{ET}, \text{LightGBM}, \text{LR}, \text{TabNet}\}$:

$$\hat{y}^{(k)} = M(X_{\text{bal}}^{(k)})$$

Evaluate on: $X_{\text{test}}^{(k)}[S_k]$

Compute: Accuracy, F1-score, AUC-ROC

- viii. Aggregate Performance Across Outer Folds

$$\mu = \frac{1}{K} \sum_{k=1}^K \text{metric}_k$$

- ix. Statistical Validation (McNemar Test)

For each configuration C vs optimal configuration C_4 :

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

- x. Final Model: Train on full balanced dataset using C_4 ; extract top-10 feature importances.

Algorithm 1. Hybrid Feature Selection with Stratified 5×3 CV

Class balancing

Maintaining class balance is essential for developing reliable and unbiased ML or DL models for churn prediction. Class balancing is the process of keeping the number of samples in each class around the same so that the predictive model may learn without favoring the class with the most examples. In churn prediction scenarios, data imbalance happens a lot since there are usually a lot more customers who stay than those who go. We use class balancing methods like synthetic minority oversampling (SMOTE) before training the classifier to fix this problem. Using SMOTE can be hard when you have to estimate churn for numerous classes with little data. This study underscores the imperative of rectifying class imbalance to guarantee that the framework generalizes effectively across all classes without yielding to overfitting. To improve the representation of minority classes, SMOTE is used to make random samples look like they are from the same group. Then, a performance evaluation is done to look for signs of overfitting or underfitting. Both under-sampling and over-sampling can fix class imbalance, but over-sampling is better because the dataset is small. Oversampling can cause overfitting; hence, it is only used on the training set and not the test set. This makes sure that the training data is fair and useful for making accurate churn predictions. The class distribution after resampling is applied to the training set, as shown in Fig. 4.

Classifiers

The framework that was proposed ensured that attrition prediction functioned effectively in a diverse array of scenarios by employing a broad selection of deep learning and machine learning classifiers. In order to obtain the optimal features for each classifier, we implemented the hybrid feature selection methodology that we previously discussed. The system utilized both traditional ensemble models and neural architectures to capitalize on the advantages of interpretability, robustness, and representation learning.

We selected ensemble-based models such as Random Forest, Extra Trees, XGBoost, and LightGBM due to their demonstrated efficiency in analyzing high-dimensional tabular data with non-linear correlations and feature interactions. Tree-based algorithms can recognize decision boundaries that are complex and use implicit feature selection during training, which makes them useful for classifying telecom churn. SVMs and LRs served as a foundation for comparative analysis because they offer generalizable linear and kernel-based decision functions.

We employed the TabNet deep learning model as it is specially designed for tabular data as opposed to other neural networks. TabNet distinguishes itself from other NNs by using decision phases with feature masking to allow comprehension of the learning process. This allows the identification of critical predictive characteristics while maintaining a high level of precision. TabNet was included to provide a comparison of deep learning and tree-based models within a single feature-optimized framework. Grid-based hyperparameter optimization was used to refine each classifier and identify the optimal parameter combinations, as shown in Table 2.

Results and discussion

This section describes the experimental setting, the utilized datasets, and the outcomes achieved by combining machine learning methods for customer churn prediction with the MHFS technique for feature selection. All datasets were pre-processed to remove samples with missing values and irrelevant characteristics before applying MLAs as discussed in Sect. 3.3. As for each dataset, the best-performing MLA(s) based on each performance metric were determined and presented in the form of tables and charts.

All experiments were conducted on a workstation equipped with an AMD Ryzen 5 3600 processor (6 cores, 12 threads), 16 GB RAM, and a 512 GB NVMe SSD, running Windows 11 (64-bit). The models were implemented in Python and related libraries. Parallel processing was enabled where applicable to utilize multi-core CPU capabilities.

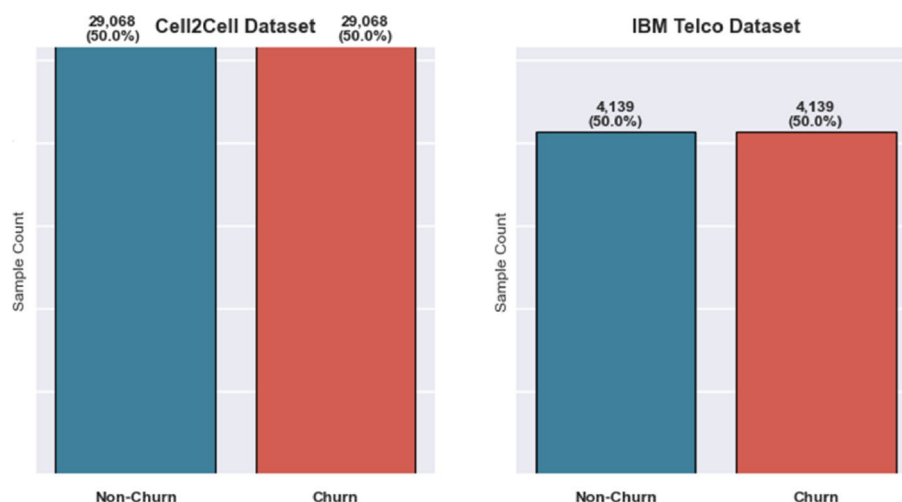


Fig. 4. Class distribution after SMOTE resampling.

Classifier	Key hyperparameters	TUNING RANGE/VALUE
Random Forest	n_estimators, max_depth, min_samples_split, criterion, class_weight	100–500; 10–50; 2–10; {gini, entropy}; {None, balanced, balanced_subsample}
Extra Trees	n_estimators, max_features, min_samples_leaf	100–400, {'auto', 'sqrt'}, 1–5
XGBoost	n_estimators, learning_rate, max_depth, subsample, colsample_bytree, reg_alpha, reg_lambda	100–500; 0.01–0.3; 3–10; 0.6–1.0; 0.6–1.0; 0–1; 0–5
LightGBM	num_leaves, learning_rate, n_estimators, min_child_samples, feature_fraction	20–200; 0.01–0.2; 100–400; 10–50; 0.6–1.0
SVM	kernel, C, gamma	{linear, rbf, poly}; 0.1–100; 0.001–1 (rbf/poly only)
Logistic Regression	penalty, solver, C, class_weight	{l1,l2,elasticnet}; {lbfgs,saga}; 0.01–100; {None, balanced}
TabNet	n_d, n_a, n_steps, gamma, lambda_sparse, learning_rate, batch_size	16–64; 16–64; 3–10; 1.0–2.0; 0.001–0.01; 0.001–0.05; 256–1024

Table 2. Hyperparameters of classifiers.

Evaluation metrics

For each algorithm, five measures were assessed to determine which performs best. Standard evaluation matrices were utilized in this research as shown in Eqs. 7, 8, 9, and 10.

Accuracy: Measures the proportion of all correct predictions.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (7)$$

Precision: Measures how the model observes the actual number of positives against the predicted positives.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (8)$$

Sensitivity (Recall): Also known as Recall, sensitivity measures the proportion of actual positives that were correctly identified.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (9)$$

AUC (Area Under the Receiver Operating Characteristic Curve): The chance that a random positive instance would be ranked higher than a random negative instance is signified by the AUC. An improved score denotes a more precise execution.

$$\text{AUC} = \frac{1}{2} - \frac{\text{FP} + \text{TP}}{2(\text{FT} + \text{TN}) + 2(\text{TP} + \text{FN})} \quad (10)$$

Nested cross-validation

To ensure unbiased model assessment and eliminate information leakage, a fully nested stratified cross-validation strategy was employed. Specifically, a fivefold outer loop was used for generalization performance estimation, while a threefold inner loop was used for feature selection validation and optimization. Within each outer training partition, all feature selection stages (Mutual Information, Recursive Feature Elimination, Boruta, and Simulated Annealing) were executed strictly on the inner training folds. The optimal feature subset was selected based on the mean inner-fold F1-score and subsequently applied to the corresponding outer test fold. No feature selection or resampling was done on the complete dataset before the split. SMOTE-based class balancing and feature standardization were applied exclusively to the outer training partitions. This 5 × 3 nested cross-validation guarantees strict separation between training and testing data to produce statistically reliable performance estimates without any data leakage.

Classification results

We utilized Cell2Cell and IBM Telco Customer Churn datasets to evaluate the efficacy of the proposed framework in forecasting customer attrition. This was done to make sure that the framework can be applied to different telecommunications contexts. The proposed Hybrid Multi-Stage Feature Selection (HMFS) method created an optimized hybrid feature set which was utilized for training of each classifier. SMOTE oversampling was then used to balance the dataset. Table 3 presents the classification metrics obtained, including Accuracy, F1-score, and AUC-ROC for both datasets, with the highest scores underlined.

XGBoost achieved the highest overall predictive performance on both datasets among the evaluated classifiers. On the Cell2Cell dataset, the model achieved an accuracy of 77.52%, an F1-score of 0.752, and an AUC-ROC of 0.854. On the IBM Telco dataset, it achieved an accuracy of 81.60%, an F1-score of 0.818, and an AUC-ROC of 0.906. The strong performance can be attributed to XGBoost's ability to model nonlinear relationships and handle class imbalance effectively through gradient boosting and regularization mechanisms.

Random Forest provided comparable performance by achieving 77.25% accuracy, 0.741 F1-score, and 0.843 AUC-ROC on Cell2Cell, and 81.45% accuracy, 0.818 F1-score, and 0.900 AUC-ROC on IBM Telco. LightGBM

Classifier	Dataset 1 cell2cell			Dataset 2 IBM telecom churn		
	Accuracy	F1	AUC-ROC	Accuracy	F1	AUC-ROC
RF	77.25%	0.741	0.843	81.45%	0.818	0.900
XGBoost	77.52%	0.752	0.854	81.60%	0.818	0.906
LightGBM	77.12%	0.751	0.848	81.42%	0.810	0.901
SVM	74.33%	0.741	0.842	81.02%	0.808	0.894
Extra Trees	75.10%	0.745	0.843	81.21%	0.812	0.897
LR	66.32%	0.670	0.755	75.05%	0.753	0.844
Tabnet	67.40%	0.673	0.750	80.14%	0.804	0.880

Table 3. Accuracy report.

Data	Stage	Accuracy without smote	Accuracy with smote	No of features	Mean FS runtime
IBM Telco	No Feature Selection	78.62%	79.67%	28	0 s
	MI	79.54%	80.53%	27	0.47 s
	MI + RFE	79.62%	80.92%	26	2.89 s
	MI + RFE + Boruta	79.68%	81.62%	24	15.44 s
	MI + RFE + Boruta + SA	79.65%	81.60%	16	46.97 s
Cell2Cell	No Feature Selection	75.10%	76.05%	57	0 s
	MI	75.75%	76.80%	40	8.80 s
	MI + RFE	75.94%	77.14%	38	70.31 s
	MI + RFE + Boruta	76.65%	77.55%	37	159.73 s
	MI + RFE + Boruta + SA	76.63%	77.52%	25	313.46 s

Table 4. Ablation study and dimensionality analysis.

produced similar competitive results with accuracies of 77.12% (Cell2Cell) and 81.42% (IBM). It confirms that the ensemble-based methods perform effectively when integrated with optimized feature subsets.

The SVM classifier achieved 74.33% accuracy and 0.741 F1-score on Cell2Cell and 81.02% accuracy with 0.808 F1-score on IBM Telco. It shows that margin-based classifiers remain competitive even after dimensionality reduction. The Extra Trees classifier provided stable and balanced results across both datasets by achieving 75.10% accuracy (Cell2Cell) and 81.21% accuracy (IBM Telco). In contrast, Logistic Regression and TabNet produced comparatively lower predictive performance on the Cell2Cell dataset. It shows that the linear models and deep tabular architectures may be more sensitive to feature characteristics and dataset distribution.

Ablation study and dimensionality analysis

Table 4 presents the ablation study conducted to justify the multi-stage hybrid design. Feature selection approaches were incrementally added. For the IBM Telco dataset, accuracy improved progressively from 79.67% (No Feature Selection) to 81.60% (Full 4-stage HMFS). Similarly, on Cell2Cell, accuracy increased from 76.05% to 77.52% under the full hybrid configuration. Even though the performance gains are incremental, the most significant contribution of the final Simulated Annealing (SA) stage lies in dimensionality reduction. For IBM Telco, the feature space was reduced from 28 initial features to 16 optimized features, while maintaining the highest predictive performance. For Cell2Cell, the feature space was reduced from 57 features to 25 optimized features, which shows competitive performance with a reduced feature subset. It shows that the hybrid framework can achieve marginal accuracy improvements while reducing the dimensionality, which is critical in real-world telecommunication deployment scenarios.

McNemar’s statistical test was used to check the significance and to compare the baseline configurations against the four-stage HMFS (MI + RFE + Boruta + SA) framework. This was done to ensure that the differences in results are not due to random variation. Since multiple pairwise comparisons were performed, a Bonferroni correction was applied, which resulted in a corrected significance level of $\alpha=0.0125$. Comparisons between the full HMFS configuration and simpler variants (0_FS, 1_MI, 2_MI_RFE, and 3_MI_RFE_Boruta) showed statistically significant differences. The McNemar test statistics ranged from 46.59 to 53.76, with corresponding p-values < 0.001 in all cases, which are well below the Bonferroni-adjusted threshold of 0.0125. It shows that the improvements observed with the complete hybrid feature selection framework are statistically significant and unlikely to be due to random chance.

The MHFS framework introduces additional computational overhead due to Boruta and SA. As reported in Table 4, the mean feature selection runtime increases progressively across configurations. For the IBM Telco dataset, runtime grows from 0.47 s (MI) to 46.9 s (MI + RFE + Boruta + SA), while for Cell2Cell it increases from 8.80 s (MI) to 313.46 s in the full hybrid configuration. This increase is expected because Boruta performs iterative wrapper-based evaluation and SA conducts stochastic search across the feature subset space. However, despite the increased preprocessing cost, the final feature dimensionality is reduced. For IBM Telco, features

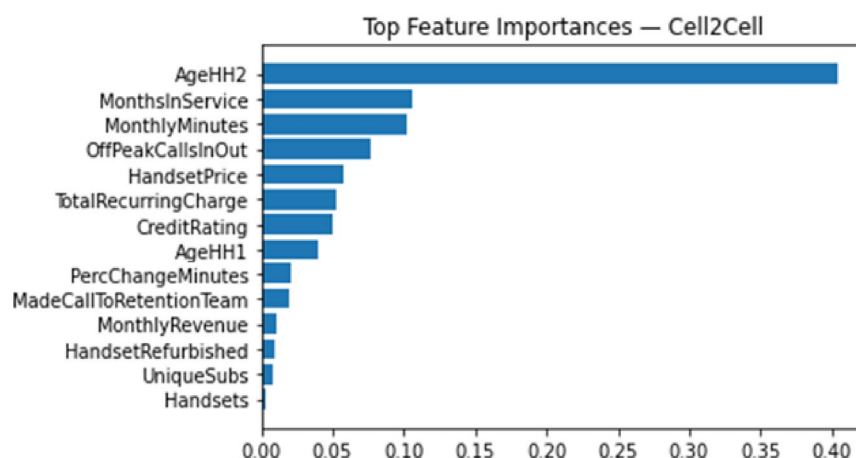


Fig. 5. Feature Importance—Cell2Cell.

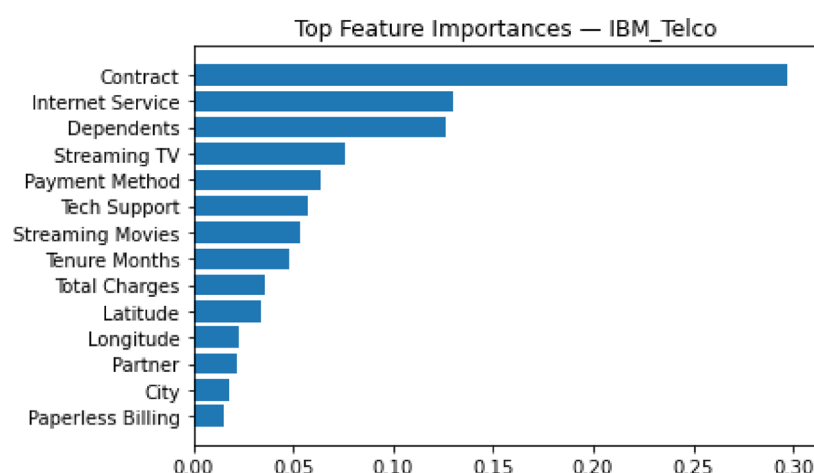


Fig. 6. Feature Importance—IMB_Telco.

are reduced from 28 to 16 (43% reduction), and for Cell2Cell from 57 to 25 (56% reduction). This reduction decreases model inference time, which is critical in churn prediction systems deployed at scale. Normally, feature selection is performed offline during model development, whereas inference is executed repeatedly in production. Therefore, the one-time computational cost is better for long-term deployment.

Discussion

This research indicates that the best results on all criteria used to predict the churn of customers cannot be achieved using ML classifiers alone. However, the performance of each individual classifier largely depends on the attributes of the given dataset and the optimized feature space provided by the proposed Hybrid Multi-Stage Feature Selection (HMFS) architecture. XGBoost was found to be the best-performing classifier, which achieved the highest performance by reaching 77.52% accuracy, 0.752 F1-score, and 0.854 AUC-ROC on the Cell2Cell dataset. It also achieved the highest performance by reaching 81.60% accuracy, 0.818 F1-score, and 0.906 AUC-ROC on the IBM Telco dataset. Random Forest and LightGBM classifiers were found to perform almost as well as the best-performing XGBoost classifier. SVM and Extra Trees were found to perform steadily but moderately. Logistic Regression and TabNet were used as a benchmark to the other classifiers, which showed a comparatively low level of performance.

Although the incorporation of the SA stage in the last stage did not improve the accuracy by a large margin, it did improve the dimensionality reduction. For example, the dimensionality of the IBM Telco data was reduced from 28 features to 16, while the Cell2Cell data reduced its dimensionality from 57 features to 25 with competitive predictive accuracy. This proves the functionality of the last stage of the HMFS algorithm as a refinement process. The McNemar's statistical testing proved that the improvements obtained through the complete HMFS pipeline were statistically significant after Bonferroni correction ($\alpha = 0.0125$), indicating that the observed differences are unlikely to be due to random variation.

Fig. 5 and Fig. 6 show the most relevant features for both datasets based on the feature importance analysis. In the Cell2Cell dataset, the most important prognostic features included AgeHH2, MonthsInService,

Ref	Method	Dataset	Accuracy	F1
37	LSTM	Cell2Cell	75.7%	0.806
38	HKD-PFLICM	Cell2Cell	76.51%	0.78
39	Transfer Learning + CNN (TL + CNN)	Cell2Cell	68.2%	-
40	CNN, LFNN	Cell2Cell	71.66%	-
41	ANN	IBM Telco	82.83%	-
42	Multi-layered ANN	IBM Telco	80.03%	-
Proposed	Hybrid feature selection + ML	Cell2Cell	77.52%	0.752
		IBM Telco	81.60%	0.818

Table 5. Comparison.

MonthlyMinutes, and OffPeakCallsInOut. In the IBM Telco dataset, the most influential features were Contract Type, Internet Service, Dependents, and Streaming TV. This correlates with previous studies that highlighted the importance of tenure, engagement, and contract types as significant telecom churn prediction factors. Although the datasets had different structures, the hybrid model successfully identified the importance of tenure and engagement features for churn prediction. From a business point of view, telecom providers can use this information to offer incentives for customers with high-tenure contracts, offer personalized services, and develop engagement strategies for high-risk customers.

While the hybridized feature selection strategy (MI + RFE + Boruta + SA) effectively optimized feature subsets, it increases computational complexity due to the iterative nature of Boruta and the stochastic search of SA. Although SA introduces controlled randomness, nested cross-validation was used to ensure stability and reduce overfitting bias. The trade-off between computational cost and dimensional efficiency should be considered when deploying the framework in large-scale industrial settings.

Comparison

The performance of the framework was assessed against proven approaches referenced in the literature. Table 5 summarizes comparative performance on the Cell2Cell and IBM Telco datasets. Deep learning architectures such as LSTM³⁷ reported a decent accuracy of 75.7%; however, such approaches typically involve higher computational cost and are sensitive to overfitting on structured tabular telecom data.

Hybrid fuzzy clustering approaches such as HKD-PFLICM³⁸ demonstrated moderate accuracy improvements but lacked cross-dataset validation. Transfer learning methods such as TL + CNN³⁹ reported lower performance on Cell2Cell (68.2%), indicating limited generalization when domain-specific churn characteristics are prominent.

Unlike many prior studies, the proposed framework employs nested cross-validation, which reduces selection bias and provides a more reliable estimate of generalization performance. Furthermore, while some previous approaches focus primarily on maximizing predictive metrics, the emphasis of this study is on hybrid feature selection for dimensionality reduction with competitive performance. The final HMFS configuration achieves 77.52% accuracy and 0.752 F1-score on Cell2Cell, and 81.60% accuracy with 0.818 F1-score on IBM Telco, while substantially reducing the feature space.

Therefore, the contribution of this work does not claim state-of-the-art performance. It provides a framework that is statistically validated and computationally analyzable for churn prediction. Furthermore, it reduces the dimensionality of the data while providing competitive performance.

Conclusion

Research on client retention is beneficial to maintain profitability in the competitive telecoms industry. This paper introduced a hybrid churn prediction framework that integrates data preprocessing, hybrid feature selection, class balancing, and ensemble machine learning within a unified pipeline. The proposed multi-stage feature selection technique combines Mutual Information (MI), Recursive Feature Elimination (RFE), Boruta, and Simulated Annealing (SA) to derive compact and informative feature subsets. Experimental evaluation on the Cell2Cell and IBM Telco datasets shows that XGBoost achieved competitive predictive performance, as it achieved the highest results with 77.52% accuracy and 0.752 F1-score on Cell2Cell, and 81.60% accuracy with 0.818 F1-score on IBM Telco. The use of 5 × 3 stratified cross-validation allowed model validation while preventing data leakage. It is important to note that the proposed hybrid pipeline substantially reduced dimensionality (e.g., 57 to 25 features in Cell2Cell and 28 to 16 in IBM Telco) while maintaining competitive predictive performance. These findings show that the primary contribution of the proposed framework lies in achieving dimensionality reduction with competitive classification performance rather than claiming superiority over all existing approaches. By integrating statistical validation and nested cross-validation, the framework provides a reproducible and computationally analyzable solution for telecom churn prediction. Future research may extend the methodology to emerging telecom environments such as 5G and IoT services and explore scalability optimizations for high-dimensional datasets.

Data availability

The publicly available datasets utilized during the current study are available from the corresponding author on request.

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Amina Saleh Omar (ASO): Conceptualization, methodology, implementation, data analysis, and original draft preparation. Kennedy Ogada (KO): Methodological refinement, supervision, and critical revision of the manuscript. Kevin Tole (KT): Validation, Computational analysis, and review and editing.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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